

● POWER BI · END-TO-END DATA ANALYTICS PROJECT

AdventureWorks

Sales & Customer Intelligence Dashboard

A 4-page interactive dashboard built from a cloud data pipeline — CSV to MySQL to a star-schema model with 20+ DAX measures.

\$24.9M

Revenue tracked

56K

Sales records

4

Report pages

20+

DAX measures

Built with Power BI · MySQL (Aiven) · Python · SQL · DAX

From raw files to a decision-ready dashboard

The goal

Recreate a professional retail analytics dashboard end to end — owning every layer from data ingestion to the final visual — using the AdventureWorks bike-shop dataset.

What it answers

- How is revenue, profit and returns trending over time?
- Which products and categories drive the business?
- Where are customers located, and who are the top spenders?
- How would a price change affect product profit?

Tech stack

Python

pandas + SQLAlchemy to load 10 CSVs to the cloud

MySQL on Aiven

Cloud database; SQL view unions 3 years of sales

Power BI

Star-schema model, Import mode, custom dark theme

DAX

Time-intelligence, TOPN, field & numeric parameters

End-to-end architecture

01

Raw CSV

10 AdventureWorks files: sales, returns, products, customers, calendar, territory



02

Cloud load

Python (pandas + SQLAlchemy) uploads to MySQL on Aiven over SSL



03

SQL model

A view UNIONs 2020–2022 sales into one 56K-row fact table



04

Power BI

Star schema, 7 relationships, 20+ DAX measures, custom theme

Data-engineering detail

Cleaned malformed export rows in-database, converted keys to integers, and added primary keys to every large table to satisfy cloud-database integrity requirements — cleaning at the source layer, not just in the BI tool.

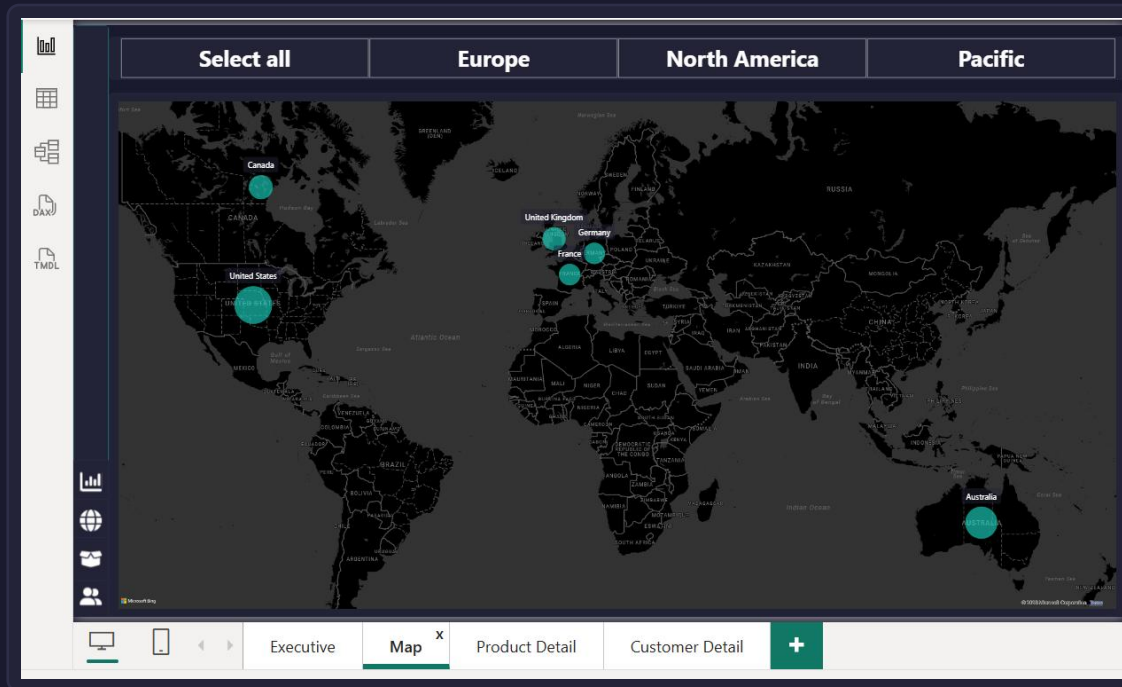
Executive Dashboard



Highlights

- KPI cards: revenue, profit, orders, return rate at a glance
- Revenue trend with trendline over 2020–2022
- Orders by category and Top 10 products table
- Return % highlighted with conditional formatting
- Monthly KPIs with month-over-month deltas

Geography & product deep-dive



Map

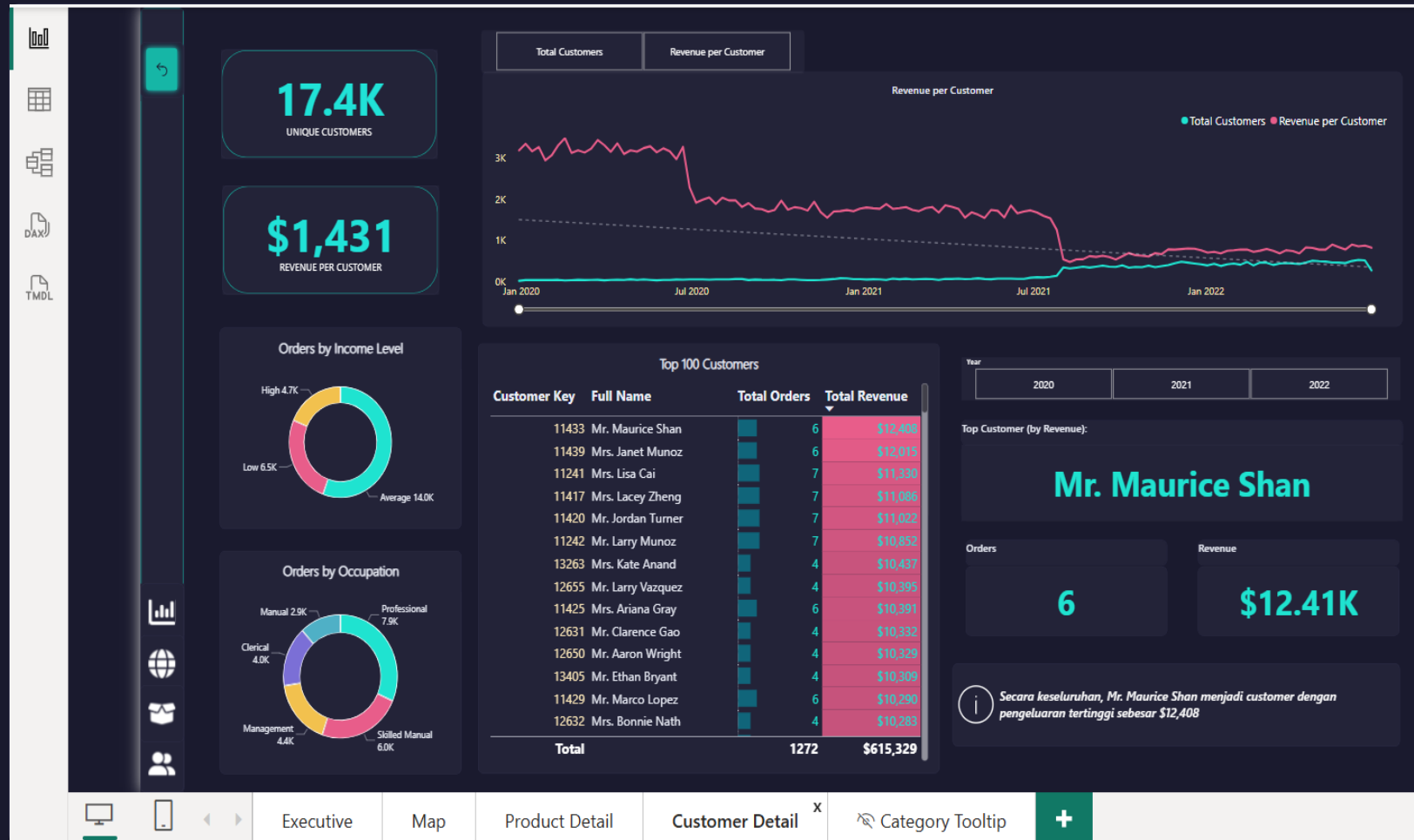
Customer distribution across regions with clickable region filters.



Product Detail

Gauges vs. target, a price-adjustment slider driving an Adjusted Profit line, and a metric selector that recolors dynamically.

Customer Detail



Highlights

- Unique customers and revenue-per-customer KPIs
- Weekly trend with zoom slider and year filter
- Top 100 customers ranked by revenue
- Income-level and occupation breakdowns
- Dynamic insight text that updates with the year filter

- UNDER THE HOOD

Technical highlights

- **Star-schema model**

One fact table linked to product, customer, territory and a marked date table via 7 relationships.

- **Time intelligence**

Revenue LY / YoY %, previous-month deltas, targets and a 90-day rolling profit measure.

- **Date hierarchy**

Custom Year → Month → Week → Date drill-down built on calculated calendar columns.

- **Field parameter**

A metric selector swaps the visualised measure and recolors the line dynamically via DAX.

- **Numeric parameter**

A price-adjustment slider feeds an Adjusted Profit measure for what-if analysis.

- **Drill-through & UX**

Cross-page drill-through, bookmark reset buttons, a slide-in slicer panel and custom tooltips.

● WHAT THIS PROJECT DEMONSTRATES

Skills demonstrated

SQL

Python

Data modeling

DAX

Data cleaning

Cloud databases

Dashboard design

Business storytelling

The takeaway

This project shows I can own an analytics workflow from raw data to a polished, interactive product — the exact skill set behind FMCG and supply-chain reporting.

Let's connect · [Full project on GitHub & Hugging Face](#)